

ECE 330 Exam 1: Spring 2022
 O. Mironenko and R. Zhang
 90 Minutes

NAME Solutions

NetID (please print) _____

USEFUL INFORMATION

$$\sin(x) = \cos(x - 90^\circ)$$

$$0 < \theta < 180^\circ \text{ (lag)}$$

$$-180^\circ < \theta < 0 \text{ (lead)}$$

$$\bar{V} = \bar{Z}\bar{I} \quad \bar{S} = \bar{V}\bar{I}^* = P + jQ$$

$$I_L = \sqrt{3}I_\phi \text{ (delta)} \quad \bar{Z}_Y = \bar{Z}_\Delta/3$$

$$V_L = \sqrt{3}V_\phi \text{ (wye)} \quad \mu_0 = 4\pi \times 10^{-7} \text{ H/m}$$

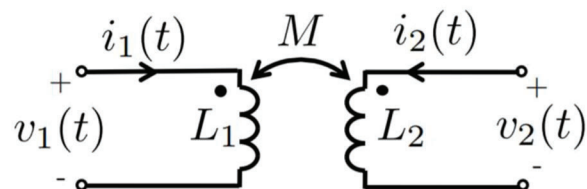
$$\bar{S}_{3\phi} = \sqrt{3}V_L I_L \angle \theta$$

ABC phase sequence has A at 0, B at -120°, and C at +120°

$$\int \underline{H} \cdot \underline{dl} = \int \underline{J_f} \cdot \hat{n} dA \quad \int \underline{E} \cdot \underline{dl} = -\frac{d}{dt} \left(\int \underline{B} \cdot \hat{n} dA \right) \quad \mathcal{R} = \frac{l}{\mu A} \quad Ni = \mathcal{R}\phi$$

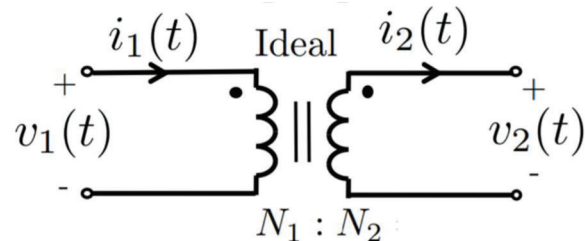
$$\phi = BA \quad \lambda = N\phi = Li \text{ (if linear)} \quad v = \frac{d\lambda}{dt} \quad k = \frac{M}{\sqrt{L_1 L_2}}$$

$$1 \text{ hp} = 746 \text{ W}$$



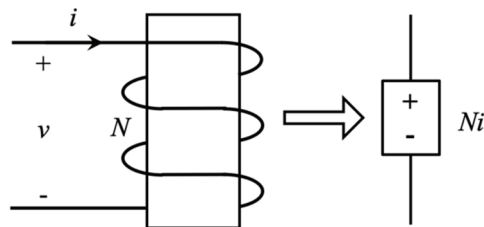
$$v_1 = L_1 \frac{di_1}{dt} + M \frac{di_2}{dt}$$

$$v_2 = M \frac{di_1}{dt} + L_2 \frac{di_2}{dt}$$



$$a = \frac{N_1}{N_2} \quad N_1 i_1 = N_2 i_2$$

$$\frac{v_1}{v_2} = \frac{N_1}{N_2}$$



Problem 1 (25 Points)

A single phase source is serving two loads connected in parallel:

- Load 1 draws 100 W at a lagging PF of 0.6.
- Load 2 draws 1 A at a leading PF of 0.6.

The voltage across the loads is $\bar{V}_{\text{load}} = 100\angle 45^\circ$ V.

You may leave complex numbers in either rectangular or polar form. Please state units.

a) Compute the current phasor $\bar{I}_1$ for Load 1.

$$\bar{I}_1 = 1.667 \angle -8.13^\circ$$

$100\angle 45^\circ \rightarrow \bar{V} \bar{I}_1^* = \frac{100}{0.6} \angle +\arccos(0.6)$

$S = \frac{100}{0.6}$

$P = 100W$

$\bar{I}_1 = \left(\frac{\frac{100}{0.6} \angle +\arccos(0.6)}{100 \angle 45^\circ} \right)^*$

$= \frac{5}{3} \angle 45^\circ - \underbrace{\arccos(0.6)}_{53.13}$

b) Compute the current phasor $\bar{I}_2$ for Load 2.

$$\bar{I}_2 = 1 \angle 98.13^\circ$$

$\bar{V} \bar{I}_2^* = 100 \angle -\arccos(0.6)$

$\bar{I}_2 = \left(\frac{100 \angle -\arccos(0.6)}{100 \angle 45^\circ} \right)^*$

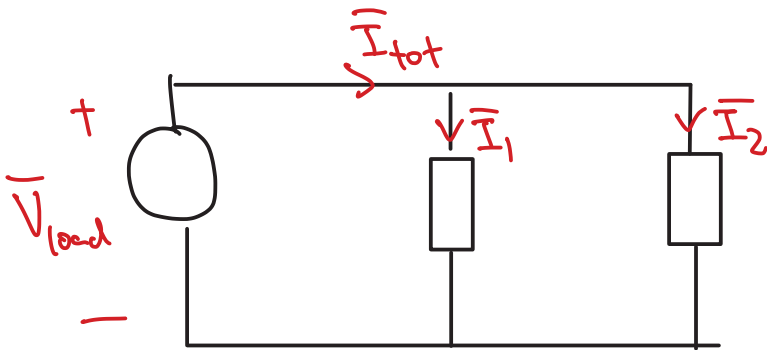
$= 1 \angle 45^\circ + \underbrace{\arccos(0.6)}_{53.13}$

$= 1 \angle 98.13^\circ$

c) What is the combined power factor over the two loads? State whether it is leading or lagging.

Combined PF = 0.948

Circle one: Leading / Lagging



$$\bar{I}_{tot} = I_{tot} \angle \theta_i$$

$$\phi_{tot} = \underbrace{\theta_v}_{45^\circ \text{ Given}} - \underbrace{\theta_i}_{26.47^\circ \text{ See below}}$$

$$= 18.53^\circ$$

Claim: $\theta_i = 26.47^\circ$

$$\bar{I}_1 = 1.667 \angle -8.13^\circ$$

$$\bar{I}_2 = 1 \angle 98.13^\circ$$

$$\cos \phi_{tot} = 0.948$$

$$\text{sign}(\phi_{tot}) > 0 \Rightarrow \text{lag}$$

$$\bar{I}_{tot} = [1.667 \cos(-8.13^\circ) + \cos(98.13^\circ)]$$

$$+ j [1.667 \sin(-8.13^\circ) + \sin(98.13^\circ)]$$

$$= 1.512 + j0.753$$

$$= 1.68913 \angle 26.47^\circ$$

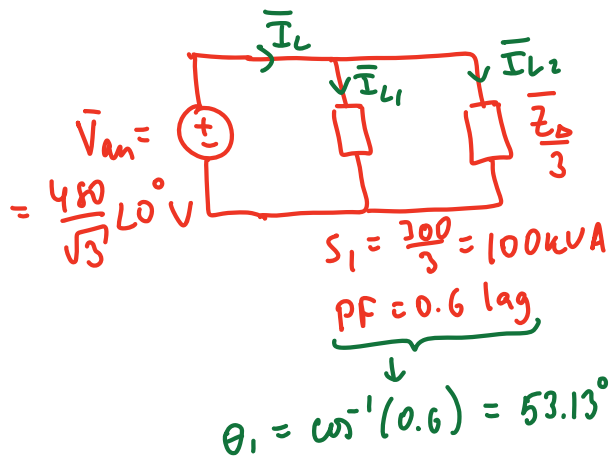
Problem 2 (25 Points)

Two 3-phase loads are connected in parallel to a wye-connected source of 480V line to line. The loads are given as:

Load 1: wye-connected, consumes 300 kVA at 0.6 lagging power factor

Load 2: delta-connected with per-phase impedance $\bar{Z}_\Delta = 50 + j50 \Omega$.

- a) Draw per-phase equivalent circuit and label:
- Source voltage phasor
 - Complex power for load 1 in terms of kVA and PF (e.g. 10 kVA at 0.1 PF leading)
 - Impedance for load 2 in terms of $\bar{Z}_\Delta$ (e.g. $\bar{Z}_\Delta / \sqrt{10}$)



$$\bar{I}_{L1}: \bar{S}_1 = \bar{V}_{an} \cdot \bar{I}_{L1}^* \Rightarrow \bar{I}_{L1} = \left(\frac{\bar{S}_1}{\bar{V}_{an}} \right)^* = \left(\frac{100 \angle 53.13^\circ}{\frac{480}{\sqrt{3}} \angle 0^\circ} \right)^* = 361 \angle -53.13^\circ \text{ A}$$

$$\bar{I}_{L2}: \bar{I}_{L2} = \frac{\bar{V}_{an}}{\frac{\bar{Z}_\Delta}{3}} = \frac{\frac{480}{\sqrt{3}} \angle 0^\circ}{\frac{70.7 \angle 45^\circ}{3}} = 11.7 \angle -45^\circ \text{ A}$$

$$\bar{I}_L = \bar{I}_{L1} + \bar{I}_{L2} = 373 \angle -53^\circ \text{ A}$$

- b) Find line current for Load 1 ($\bar{I}_{L1}$), line current for Load 2 ($\bar{I}_{L2}$), and the total source line current $\bar{I}_L$. Express all currents in a polar form (magnitude and angle). (Use next page for work. Please state units)

$$\bar{I}_{L1} = 361 \angle -53.13^\circ \text{ A}$$

$$\bar{I}_{L2} = 11.7 \angle -45^\circ \text{ A}$$

$$\bar{I}_L = 373 \angle -53^\circ \text{ A}$$

- c) A delta connected capacitor bank is added in parallel to the loads to bring the overall power factor to unity. What is the new line current? Express the current in a polar form (magnitude and angle). Please state units.

$$\overline{I_{Lnew}} = \underline{223.8 \angle 0^\circ A}$$

Working per-phase:

$$\overline{S}_{tnew} = \overline{V}_{an} \cdot \overline{I}_{Lnew}^* \Rightarrow \overline{I}_{Lnew} = \left(\frac{\overline{S}_{tnew}}{\overline{V}_{an}} \right)^*$$

$$PF_{new} = 1 \Rightarrow Q_{new} = 0 \Rightarrow \overline{S}_{tnew} = P + j0$$

$$\overline{S}_{told} = P + jQ_{old}$$

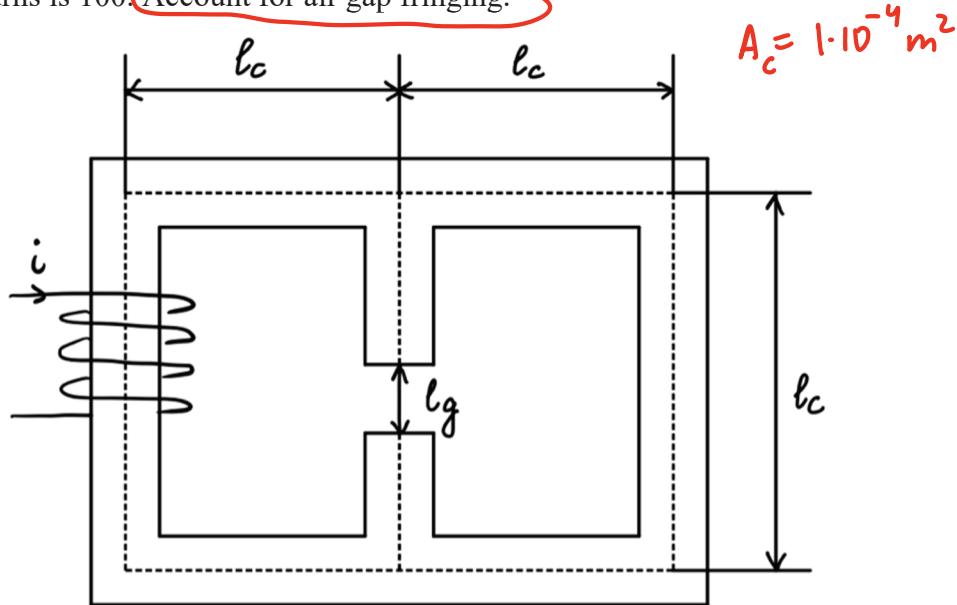
$$\begin{aligned} \overline{S}_{told} &= \overline{V}_{an} \cdot \overline{I}_L^* = \frac{480}{\sqrt{3}} \angle 0^\circ \cdot (373 \angle -53^\circ)^* = 103 \angle 53^\circ = \\ &= \underbrace{62}_{P} + j \underbrace{82}_{Q_{old}} \text{ kVA} \end{aligned}$$

$$\overline{S}_{tnew} = 62 + j0 \text{ kVA}$$

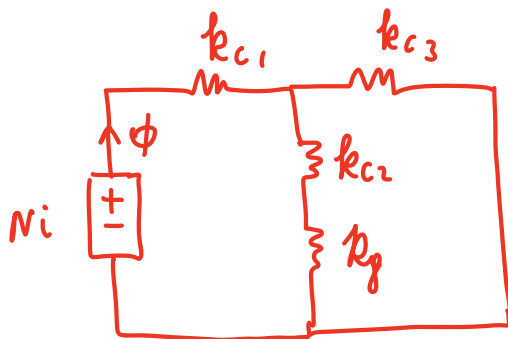
$$\overline{I}_{Lnew} = \left(\frac{62 \text{ k} \angle 0^\circ}{\frac{480}{\sqrt{3}} \angle 0^\circ} \right)^* = 223.8 \angle 0^\circ A$$

Problem 3 (25 Points)

Consider the iron core with air gap given in the figure below. Assume μ_r of the iron core = 1000, $l_c = 10$ cm, $l_g = 0.1$ cm, the cross section of the core is 1 cm by 1 cm, and the number of turns is 100. Account for air gap fringing.



- a) Draw the equivalent magnetic circuit and label direction of magnetic flux in the left leg, polarity of MMF source and all reluctances.



$$k_{c1} = \frac{2l_c}{\mu_r \mu_0 A_c} = \frac{0.3}{1000 \cdot 4\pi \cdot 10^{-7} \cdot 10^{-4}} = 2.38 \cdot 10^6 \frac{A \cdot t}{Wb}$$

$$k_{c2} = \frac{l_c - l_g}{\mu_r \mu_0 A_c} = \frac{(0.1 - 0.001)}{1000 \cdot 4\pi \cdot 10^{-7} \cdot 10^{-4}} = 7.87 \cdot 10^5 \frac{A \cdot t}{Wb}$$

$$k_{c3} = \frac{3l_c}{\mu_r \mu_0 A_c} = k_{c1}$$

$$k_g = \frac{l_g}{\mu_0 A_g} = \frac{0.001}{4\pi \cdot 10^{-7} \cdot 1.21 \cdot 10^{-4}} = 6.57 \cdot 10^6 \frac{A \cdot t}{Wb}$$

Fringing:

$$A_g = (1 + 0.1)(1 + 0.1) = 1.21 \cdot 10^{-4} m^2$$

b) Calculate the inductance of the coil. Please state units.

$$L = \underline{2.38 \text{ mH}}$$

$$\lambda = N\phi = Li$$

$$\phi = \frac{Ni}{k_t}$$

$$\lambda = \frac{N^2}{k_t} \cdot i$$

$$L = \frac{100^2}{4.19 \cdot 10^6} = 2.38 \text{ mH}$$

$$k_t = k_{c1} + \left((k_{c2} + k_g) \parallel k_{c3} \right) =$$
$$= 4.19 \cdot 10^6 \frac{\text{A} \cdot \text{t}}{\text{wb}}$$

c) What is the current i for a flux density $B = 0.5\text{T}$ in the left leg? Please state units.

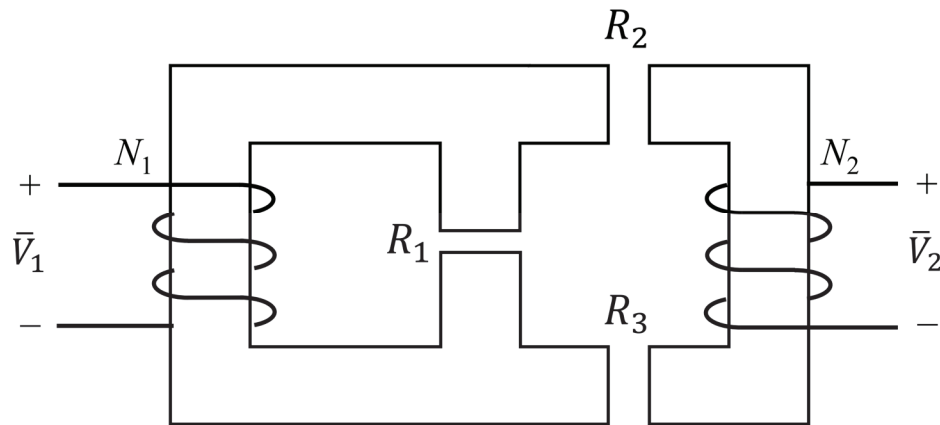
$$i = \underline{2.095 \text{ A}}$$

$$\phi = \frac{Ni}{k_t} \Rightarrow i = \frac{\phi \cdot k_t}{N} = 2.095 \text{ A}$$

$$\phi = B \cdot A_c = 0.5 \cdot 10^{-4} \text{ wb}$$

Problem 4 (25 Points)

Your local utility company is suing their general contractor for a misinstalled high-voltage transformer, and you have been asked to serve as an expert witness on the legal case. The transformer has the following schematics.



The two coils are wound to have $N_1 = 400$ turns and $N_2 = 10$ turns respectively. The three air gaps are sized to have reluctances of $R_1 = 10 \text{ At/Wb}$ and $R_2 = 500 \text{ At/Wb}$ and $R_3 = 500 \text{ At/Wb}$ as labeled. You may treat the permeability of the magnetic core as infinite.

- a) Compute the self inductances L_1 , L_2 , and the mutual inductance M of the two coils. (Use the next page for work. Please state units.)

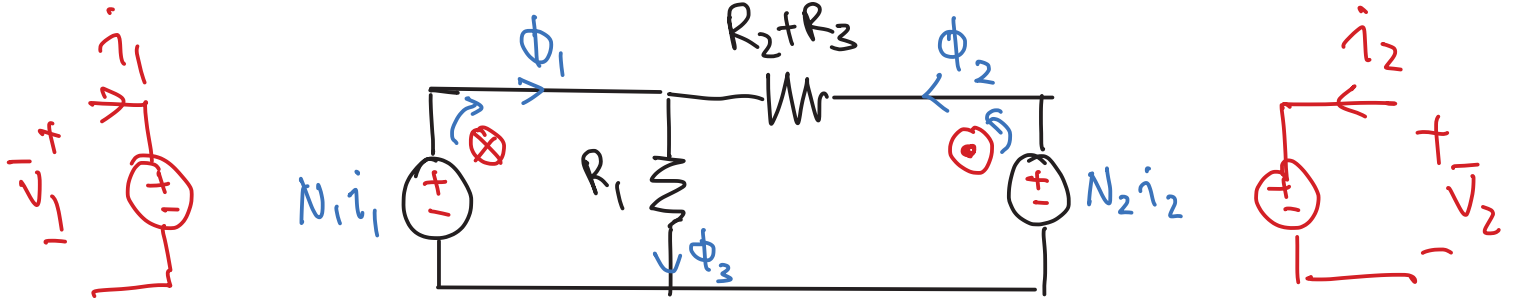
$$L_1 = \underline{16160 \text{ H}}$$

$$L_2 = \underline{0.1 \text{ H}}$$

$$M = \underline{4 \text{ H}}$$

(see next page for work)

(Blank page for Problem 4a)



$$\phi_2 = \frac{N_2 i_2 - N_1 i_1}{R_2 + R_3} \Rightarrow \lambda_2 = \frac{N_2^2}{R_2 + R_3} i_2 - \frac{N_1 N_2}{R_2 + R_3} i_1$$

$$\phi_1 + \phi_2 = \phi_3, \quad \lambda_1 = N_1^2 \left(\frac{1}{R_1} + \frac{1}{R_2 + R_3} \right) i_1 - \frac{N_1 N_2}{R_2 + R_3} i_2$$

$$\phi_1 + \frac{N_2 i_2 - N_1 i_1}{R_2 + R_3} = \frac{N_1 i_1}{R_1}$$

$$N_1 = 400, \quad N_2 = 10, \quad R_1 = 10, \quad R_2 = 500$$

$$L_2 = \frac{N_2^2}{R_2 + R_3} = \frac{100}{1 \times 10^3} = 0.1 \text{ H}$$

$$M = \frac{N_1 N_2}{R_2 + R_3} = \frac{4 \times 10^3}{1 \times 10^3} = 4 \text{ H}$$

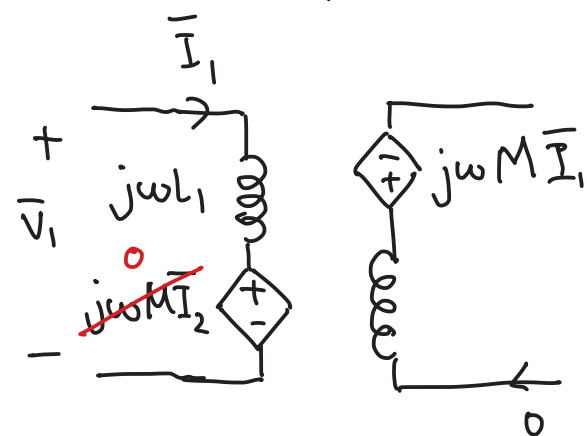
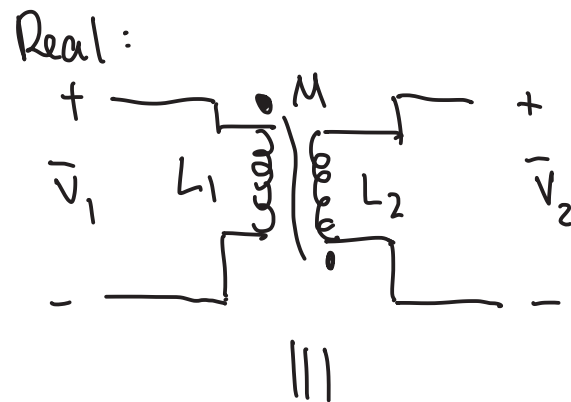
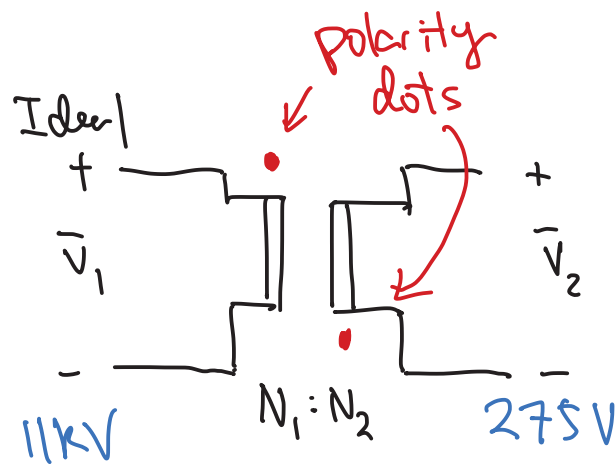
$$L_1 = \frac{N_1^2}{R_1 \parallel (R_2 + R_3)} = \frac{16 \times 10^4}{1000/101} = 16160 \text{ H}$$

b) The utility company is suing the contractor because the transformer is supposed to have an output voltage of 110V at open circuit (magnitude, rms). Using the self and mutual inductance values you have just computed, what is the actual output voltage $\bar{V}_2$ at open circuit with an input voltage $\bar{V}_1 = 11 \angle 45^\circ$ kV (rms) at 60 Hz?

Please specify both the phase and the magnitude of the output voltage $\bar{V}_2$, noting that the magnitude must be a nonnegative number. Please state units.

Magnitude = 2.7227

Phase = -135°



$$\bar{I}_1 = \bar{V}_1 / j\omega L_1$$

$$\bar{V}_2 = -j\omega M \bar{I}_1$$

$$= -\frac{j\omega M}{j\omega L_1} \bar{V}_1$$

$$= -\frac{4}{16160} 11 \times 10^3 \angle 45^\circ = 2.7227 \angle -135^\circ \text{ V}$$